MAL Omics — Proteomics Analysis Report

COPDGene — SomaScan (∼4,979 proteins) — $n = 1{,}306$

Goal. (1) Understand biological correlates of MAL detectable in peripheral plasma. (2) Develop a blood-based proteomic risk score to enrich studies for MAL and emphysema progressors.

Outcome. Emphysema progression = change in CT lung density Phase 3 − Phase 2 (`Change_lung_density_vnb_P2_P3`, HU). Negative = worsened.

Data Sources and Variable Names

Four CSV files were used. All joins are on `sid`.

`data/csv_exports/prot.csv` — SomaScan proteomics

Column	Description
`sid`	Subject identifier (join key)
`X{seqID}`	Raw RFU value for each protein (e.g. `X10000.28`). ∼4,979 columns matching pattern `^X[0-9]`. Renamed to `log2_X{seqID}` after $\log_2$ transform.

`data/csv_exports/pheno_MAL.csv` — Phenotypes and covariates

Column	Used as	Status
`sid`	Join key	Found
`Change_lung_density_vnb_P2_P3`	Outcome: emphysema progression (HU)	Found
`lung_density_vnb_P2`	Baseline lung density (HU)	Found
`Age_P2`	Age (years)	Found
`gender`	Sex (coded 1/2)	Found
`race`	Race (coded 1/2)	Found
`ATS_PackYears_P2`	Pack-years of smoking	Found
`SmokCigNow_P2`	Current smoking (0/1)	Found
`scanner_model_clean_P2`	CT scanner model	Found
`FEV1_FVC_post_P2`	FEV_1/FVC ratio	Found
`PC1` – `PC5`	Ancestry principal components	Found
`FEV1pp_P2`	FEV_1% predicted	NOT FOUND
`BMI` / `bmi_P2`	Body mass index	NOT FOUND

`data/Data/MAL.csv` — MAL scores

Column	Description
`sid`	Subject identifier (join key)
`meanmal`	MAL score (continuous, range 0.01–0.40)

`data/csv_exports/metadat5k.csv` — **Protein annotation**

Column	Description
`seqID`	Protein sequence identifier; joins to `log2_X{seqID}`
`EntrezGeneSymbol`	HGNC gene symbol (e.g. `ADIPOQ`)
`Target`	Short protein name (e.g. `Adiponectin`)
`TargetFullName`	Full protein name

Join logic. Three-way inner join: `prot` $\bowtie_{\texttt{sid}}$ `pheno` $\bowtie_{\texttt{sid}}$ `MAL`. Metadata joined on `seqID = sub("^log2_X", "", predictor)`. Starting $n = 4{,}630$ after merge; $n = 1{,}306$ after complete-case filter on all required columns.

Table 1 — Demographics

Study population after 3-way merge (proteomics × phenotype × MAL) and complete-case filtering.

Variables not found in the phenotype file:

- *Not found: FEV1pp_P2 (FEV$_1$% predicted)* — column not present under any searched pattern (`FEV1pp`, `FEV1pct`, `FEV1_percent_pred`). Row left blank.
- *Not found: BMI* — no column matching `^[Bb][Mm][Ii]` found in the phenotype file. Row left blank.

Table 1. Study population ($n = 1{,}306$). Mean (SD) for continuous; n (%) for categorical. Emphysema progression = $\rho_{\text{P3}} - \rho_{\text{P2}}$ (HU).

Characteristic	Overall ($n = 1{,}306$)
Demographics	
Age (years)	65 (8)
Sex: male / female	633 (48%) / 673 (52%)
Race: White / Black	991 (76%) / 315 (24%)
Smoking	
Current smoker	445 (34%)
Pack-years	43 (23)
Pulmonary function	
FEV$_1$% predicted	*Not found: column not found in phenotype file*
FEV$_1$/FVC	0.69 (0.14)
CT imaging	
Lung density Phase 2 (HU)	83 (23)
MAL score	0.04 (0.05)
Emphysema progression (HU)	1 (7)
Anthropometrics	
BMI (kg/m^2)	*Not found: column not found in phenotype file*

Note. Sex is coded 1/2 in the data; assumed Male/Female per COPDGene convention. Race coded

1/2; assumed White/Black. Emphysema progression mean of 1 HU (SD 7) reflects a near-zero mean — the distribution includes both progressors (negative $\Delta\rho$) and improvers (positive $\Delta\rho$).

Figure 1 — Study Design

Not automatically generated. Requires a manual schematic (e.g., BioRender) illustrating the analysis flow: SomaScan proteomics → differential expression → GSEA / STRING → adaptive LASSO PRS → emphysema progression association → causal mediation.

[Figure 1 placeholder — study design schematic to be drawn manually, e.g., BioRender]

Figure 1: Study design schematic.

Figure 2 — Protein Expression Landscape of MAL

A. Volcano Plot (protein $\sim$ MAL)

What we are looking at. A volcano plot shows which proteins in the blood are statistically associated with MAL (Mechanically Affected Lung score, range 0.01–0.40; higher = more lung under destructive mechanical stress). Each dot is one protein.

- **x-axis ($\hat{\beta}$):** how much that protein changes per one-unit increase in MAL, after adjusting for age, sex, race, smoking, pack-years, scanner, and ancestry PCs. A *positive* $\hat{\beta}$ means the protein is *higher* in people with more mechanical lung stress. A *negative* $\hat{\beta}$ means it is *lower*. MAL ranges from 0.01 to 0.40, so a $\hat{\beta}$ of 0.27 (Adiponectin) means moving across the full MAL range would correspond to a $0.27 \times 0.39 \approx 0.1$ $\log_2$-unit change — a modest but detectable shift.
- **y-axis ($-\log_{10}$ FDR):** confidence. Higher = more significant. Dots above the horizontal threshold (FDR < 0.05) are coloured: red = enriched, blue = depleted. Grey dots did not reach significance.
- **A dot far right and high up** is a protein that is both strongly and reliably elevated in subjects with more MAL.
- **A dot far left and high up** is a protein that is both strongly and reliably lower in subjects with more MAL.

How it was calculated. For each of 4,979 proteins, the model is:

$$\log_2(\text{protein}) \sim \beta_{\text{MAL}} \cdot \texttt{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{scanner} + \text{PC1–PC5}$$

$\hat{\beta}_{\mathrm{MAL}}$ tells us: for a one-unit increase in MAL, how much does the $\log_2$ protein level change, after adjusting for the listed covariates? FDR correction (Benjamini–Hochberg) across all 4,979 tests.

Each point is one protein. x-axis: MAL $\hat{\beta}$. y-axis: $-\log_{10}(\mathrm{FDR})$. Red = FDR < 0.05 and enriched; blue = FDR < 0.05 and depleted; grey = not significant. Top 20 by FDR are labelled.

Key result. 45 of 4,979 proteins were significant at FDR < 0.05.

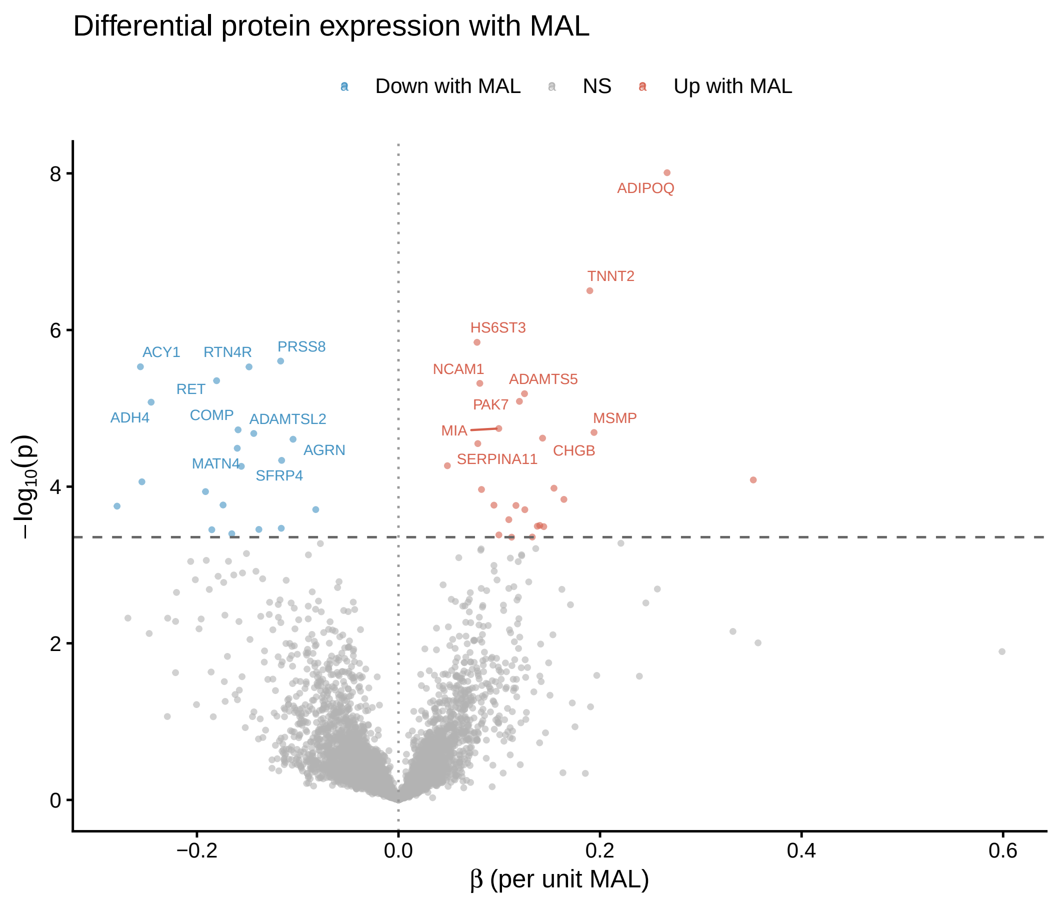


Figure 2: Volcano plot: differential protein expression with MAL (n = 4,979 proteins). Red = enriched, blue = depleted at FDR < 0.05. Top 20 proteins labelled.

Top enriched (higher with more MAL): *ADIPOQ* (Adiponectin, $\beta = 0.27$), *TNNT2* (Troponin T, $\beta = 0.19$), *ADAMTS5*, *NCAM1*, *PAK7*.

Top depleted (lower with more MAL): *ACY1* ($\beta = -0.26$), *ADH4* ($\beta = -0.25$), *RTN4R* (Nogo Receptor), *RET*, *PRSS8*, *SFRP4*.

B. GSEA Barplot

What we are looking at. Instead of asking "is this one protein associated with MAL?", GSEA asks: "does this entire biological *pathway* show a coordinated shift with MAL?" A pathway might have no single dramatically changed protein, but if most of its member proteins are all slightly higher in high-MAL subjects, that is a real biological signal.

Each bar in the plot is one significant pathway.

- **Bar pointing right (NES > 0, enriched):** the pathway's proteins are, on average, *higher* in plasma when MAL is higher. This pathway is more active in people with more mechanically stressed lung.
- **Bar pointing left (NES < 0, depleted):** the pathway's proteins are *lower* in plasma when MAL is higher. This pathway is suppressed in people with more mechanical stress.
- **Longer bar = stronger shift.** NES of ± 2 is a strong, consistent signal across the gene set.
- **padj:** how confident we are the shift is real, after correcting for all pathways tested. Only bars with padj < 0.05 are shown.

What the results mean here: seven metabolic pathways (energy, fat, amino acids) are depleted — meaning people with more MAL have lower levels of proteins involved in normal metabolism. Two immune pathways (chemokines, antimicrobial peptides) are enriched — meaning the immune system is more active. Together this suggests mechanical lung stress is associated with a body-wide metabolic slowdown and immune activation, detectable in blood.

How it was calculated. All 4,979 proteins were ranked by their MAL t-statistic (most positively associated with MAL at the top, most negatively associated at the bottom). Gene sets from three collections were tested: Hallmark, KEGG Legacy, and Reactome (gene-set sizes 15–500).

NES (Normalized Enrichment Score): measures whether the members of a gene set are concentrated at the top or bottom of the ranked protein list.

- NES > 0: the pathway's proteins tend to be *higher* in subjects with more MAL (enriched).
- NES < 0: the pathway's proteins tend to be *lower* in subjects with more MAL (depleted).
- The normalization accounts for gene-set size so scores are comparable across pathways.

padj: Benjamini–Hochberg-adjusted p-value across all pathways tested within each collection. Threshold used: padj < 0.05.

What we are seeing: 9 pathways reached significance. Seven are depleted (NES < 0) — meaning proteins in these pathways are systemically lower in people with higher MAL. All seven are core metabolic pathways: energy production (oxidative phosphorylation, glycolysis), fat breakdown (fatty acid metabolism), amino acid handling, and drug/xenobiotic processing. Two pathways are enriched (NES > 0): chemokine receptor signalling and antimicrobial peptides, both inflammatory/immune in nature.

In plain terms: higher MAL is associated with a systemic shift away from normal metabolic housekeeping and toward an inflammatory state. This is detectable in peripheral blood even though MAL is a lung structural measure.

GSEA significant pathways (padj < 0.05). NES > 0 = enriched in high MAL; NES < 0 = depleted.

Collection	Pathway	NES	padj
Hallmark	Fatty Acid Metabolism	−1.76	0.021
Hallmark	Oxidative Phosphorylation	−1.78	0.021
KEGG	Glycolysis / Gluconeogenesis	−2.02	0.018
Reactome	Biological Oxidations	−2.05	0.007
Reactome	Metabolism of Amino Acids & Derivatives	−1.84	0.013
Reactome	Diseases of Metabolism	−1.76	0.032
Reactome	Phase I Functionalization of Compounds	−1.97	0.037
Reactome	Chemokine Receptors Bind Chemokines	+2.02	0.024
Reactome	Antimicrobial Peptides	+1.94	0.032

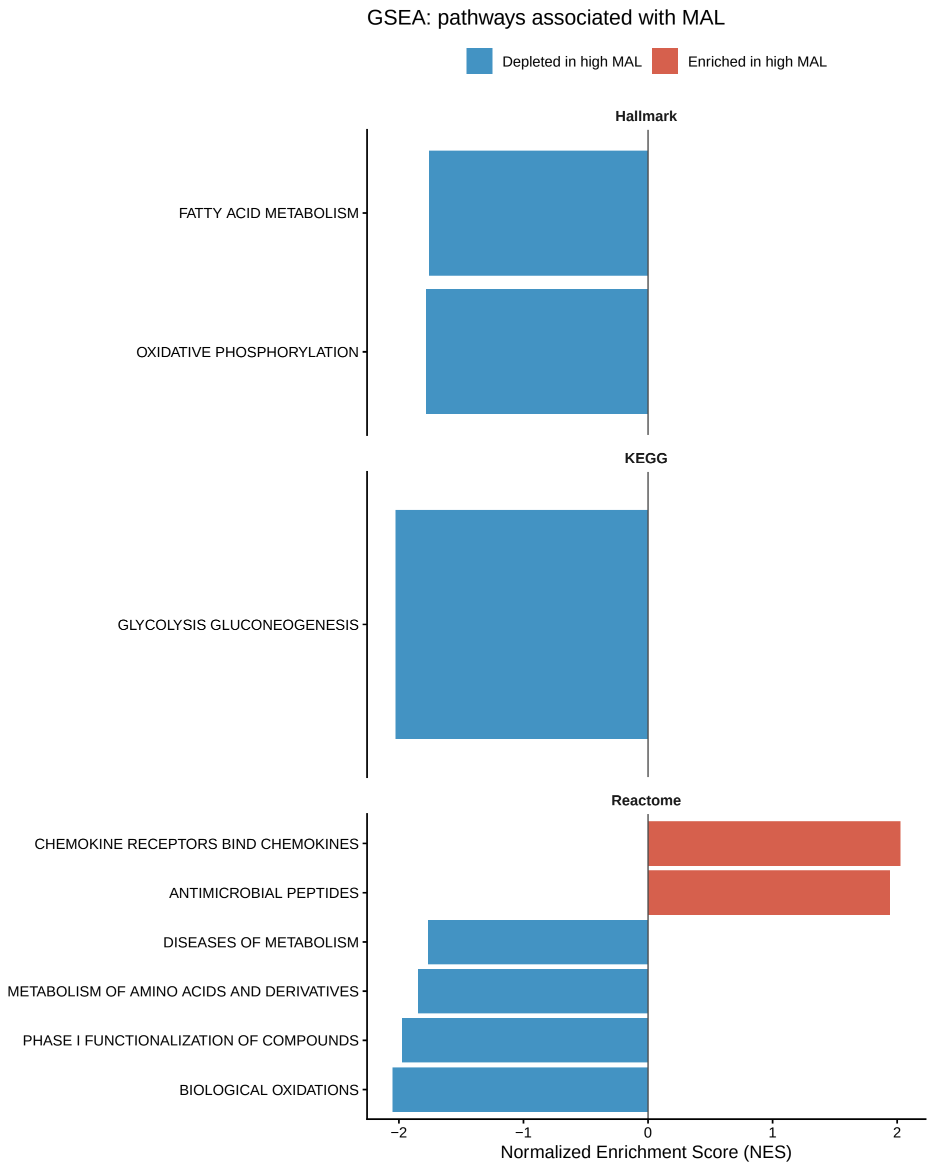


Figure 3: GSEA barplot: 9 significant pathways (padj < 0.05) faceted by collection. NES $< 0 =$ depleted in high-MAL subjects.

C. STRING Network

What we are looking at. A protein–protein interaction network for the 45 MAL-associated proteins. Each node is one protein. An edge (line) between two proteins means they are known to physically interact or co-function in humans, based on published data in the STRING database.

- **A cluster of connected nodes** means those proteins work together as part of the same biological process. If they all go up or down with MAL, that cluster represents a coherent pathway being affected — stronger evidence than if each protein were isolated.
- **A node with many edges (hub)** is a protein that interacts with many others, suggesting it may be a central regulator in the MAL response.

- **An isolated node** (no edges) means that protein had no known high-confidence interactions with other significant proteins in this set. It may still be biologically relevant, just acting independently.
- **Edge thickness/colour** reflects interaction confidence score (≥ 400 = medium confidence).

How it was calculated. The 45 FDR-significant proteins were submitted to the STRING v11.5 human database. Only interactions with a confidence score ≥ 400 (medium confidence) were drawn. Only PNG available — `STRINGdb::plot_network()` does not support PDF output.

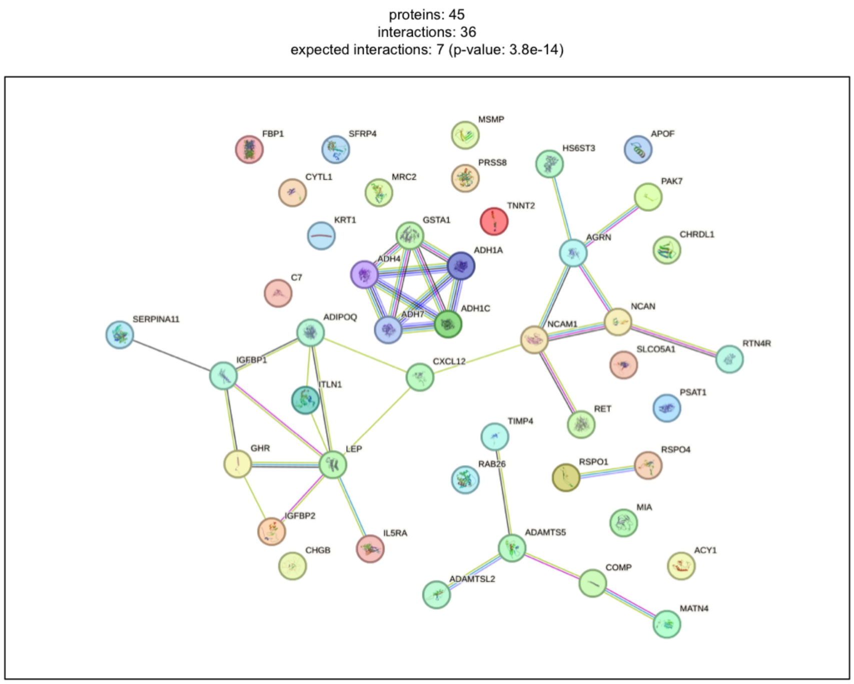


Figure 4: STRING protein–protein interaction network for 45 FDR-significant proteins (v11.5, score ≥ 400).

Figure 3 — Quartile Plot of Proteomic Risk Score

What we are looking at. The subjects are divided into four groups by their Proteomic Risk Score (quartile 1 = lowest 25% of PRS, quartile 4 = highest 25%). Within each group, we ask: does a higher PRS within that group predict more emphysema progression?

- ***x*-axis:** risk score quartile (1–4). Quartile 1 = people with the least MAL-like protein profile in blood. Quartile 4 = people with the most MAL-like protein profile.
- ***y*-axis ($\hat{\beta}$, HU/SD):** the regression coefficient for PRS in that quartile subgroup. A positive value means higher PRS within the quartile is associated with *less* negative progression (density loss smaller). A negative value means higher PRS is associated with *more* progression (more density loss). Zero means no association.

- **Error bars:** 95% confidence interval. If the bar crosses the dashed zero line, the association is not statistically significant in that quartile.
- **What a consistent upward or downward trend across quartiles would mean:** the PRS is linearly tracking disease severity. If the bars are flat and all cross zero, the score is not predicting progression at any level of risk.

Key result. Effect sizes are small and confidence intervals cross zero in all quartiles — the PRS does not predict emphysema progression within any risk stratum ($p = 0.079$ overall).

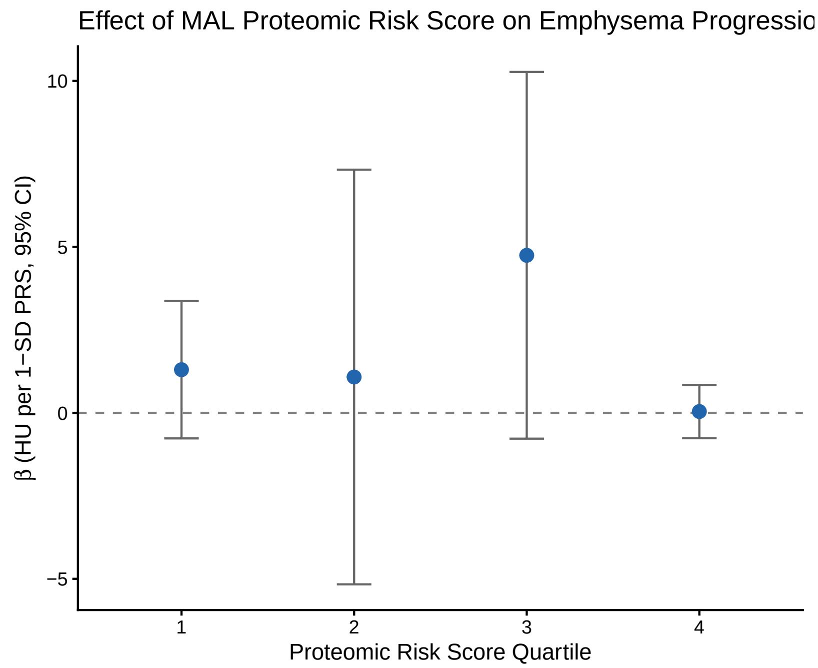


Figure 5: Quartile plot of the Proteomic Risk Score (PRS). Each point is the OLS $\hat{\beta}$ (HU/SD) for the PRS within that quartile subgroup, with 95% CI. Dashed line at zero.

Proteomic Risk Score — Methods and Results

What we are looking at. A single number per person — their Proteomic Risk Score (PRS) — computed from a weighted combination of 378 plasma proteins.

- **High PRS:** the person's blood protein pattern looks similar to someone with high MAL (more mechanically stressed lung).
- **Low PRS:** their protein pattern looks like someone with low MAL (less mechanical stress).
- **Test R^2 = 0.157:** the PRS explains 15.7% of the variation in MAL in held-out subjects. That means 84% of what makes MAL high or low is *not* captured by the blood proteome — a meaningful but modest signal.
- **Positive $\hat{\beta}$ in the weights table:** that protein is higher in people with more MAL, and contributes positively to the score.
- **Negative $\hat{\beta}$:** that protein is lower in people with more MAL, and pulls the score down.

How it was calculated. Data split. 70/30 train/test (`set.seed(2024)`), $n_{\text{train}} = 914$, $n_{\text{test}} = 392$.

Step 1 — Log2 transform all protein RFU values:

$$\tilde{x} = \log_2\bigl(\max(x,\ 10^{-6})\bigr)$$

Transforms skewed RFU values to an approximately normal scale. The 10^{-6} floor prevents log(0) errors.

Step 2 — Adaptive weights from ridge regression:

$$w_j = \frac{1}{|\hat{\beta}_j^{\text{ridge}}|}$$

A ridge regression is first fit on the residualised MAL outcome (after projecting out the unpenalised covariates). Each protein j gets a weight inversely proportional to its ridge coefficient. Proteins that ridge already identified as informative get *lower* penalty in the LASSO step; irrelevant proteins get *higher* penalty. This gives the adaptive LASSO its oracle property — it tends to select the true model.

Step 3 — Adaptive LASSO selection:

$$\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_j w_j |\beta_j|$$

λ is chosen by 10-fold cross-validation (λ_{min}). Covariates (age, sex, race, smoking, pack-years, scanner, PC1–PC5) are assigned $w_j = 0$ so they are never penalised.

Step 4 — Post-selection OLS refit. The selected proteins are refit by ordinary least squares. LASSO shrinks coefficients toward zero, so the LASSO $\hat{\beta}$ values are biased. OLS on the selected set gives unbiased estimates, proper standard errors, confidence intervals, and p-values.

Step 5 — Compute the risk score for each subject:

$$\text{PRS}_i = \sum_{j \in \mathcal{S}} \hat{\beta}_j^{\text{OLS}} \cdot \tilde{x}_{ij}$$

where $\mathcal{S}$ is the selected protein set. The PRS is then standardised to mean 0, SD 1 across the full cohort so that the regression coefficient has units of HU per SD of the score.

Result. 378 proteins selected. Test $R^2 = 0.157$.

Top 20 PRS proteins by $|\hat{\beta}|$ (OLS).

Gene	Target	$\hat{\beta}$	95% CI (low)	95% CI (high)	p
ZPBP	ZPBP1	+0.286	0.160	0.412	< 0.001
CFI	Factor I	−0.226	−0.398	−0.054	0.010
GREM2	GREM2	+0.193	0.113	0.273	< 0.001
C7	Complement C7	+0.169	0.030	0.309	0.017
SEC13	SEC13	−0.153	−0.254	−0.053	0.003
SERPINF2	α_2-Antiplasmin	+0.149	0.011	0.287	0.035
FSTL1	FSTL1	−0.147	−0.309	0.015	0.075
APOH	β_2-Glycoprotein I	+0.143	−0.049	0.335	0.145
COL3A1	Collagen Type III	+0.132	0.061	0.204	< 0.001
C3	C3a	+0.131	0.021	0.242	0.020
SMS	Spermine Synthase	−0.125	−0.179	−0.072	< 0.001
MATN2	MATN2	−0.125	−0.242	−0.008	0.036
GALNT11	GLT11	−0.122	−0.225	−0.019	0.021
ROBO2	ROBO2	−0.119	−0.220	−0.017	0.022
SEZ6L	SEZ6L	+0.117	0.020	0.214	0.019
FBLN1	Fibulin-1	−0.113	−0.278	0.051	0.175
ADAMTS5	ADAMTS-5	+0.113	0.062	0.163	< 0.001
CCL23	MPIF-1	−0.112	−0.160	−0.063	< 0.001
CREB3L4	CR3L4	+0.108	0.047	0.169	< 0.001
CNTN1	Contactin-1	−0.107	−0.229	0.015	0.086

AIC Comparison — PRS vs. Clinical Model

What we are looking at. Does the blood protein score add anything useful for predicting emphysema progression, on top of what clinical variables already tell us? Three models are compared:

- **Clinical model:** uses only age, sex, race, smoking status, pack-years, and CT scanner type. This is the baseline — what a doctor already knows.
- **PRS-only model:** uses only the protein score, baseline lung density, and scanner. No clinical variables.
- **Clinical + PRS model:** adds the protein score on top of the clinical model. This is the key test — does blood proteomics help beyond standard care variables?

Reading the AIC table:

- **Lower AIC = better model** (fits the data well without using unnecessary parameters).
- **ΔAIC vs. Clinical = 0:** the clinical model is the reference. All other models are compared to it.
- **ΔAIC > 0:** that model is *worse* than clinical alone — it either fits poorly or uses too many parameters for the gain.
- **ΔAIC ≤ -2:** that model is meaningfully *better* than clinical alone. This is the threshold we need to cross to claim the PRS adds value.
- **PRS $\hat{\beta} = 0.42$ HU/SD, $p = 0.079$:** for each 1 SD increase in PRS in the combined model, emphysema progression changes by 0.42 HU on average. Since normal progression SD is $\sim$7 HU, this is a small, non-significant effect.

How it was calculated. AIC (Akaike Information Criterion) measures model fit minus a penalty for the number of parameters. A lower AIC is better. A difference of -2 or more (i.e. the new model is at least 2 units lower) is considered a meaningful improvement. Three OLS models were fitted for emphysema progression. AIC improvement of ≤ -2 considered meaningful.

Note: BMI was not available in the phenotype file and is absent from the clinical model. This should be revisited when BMI data are available.

AIC comparison.

Model	AIC	ΔAIC vs. Clinical
Clinical (age, sex, race, smoking, pack-years, scanner)	8,815	0
PRS only (PRS + baseline density + scanner)	8,848	+33.1
Clinical + PRS	8,816	+0.8

PRS $\hat{\beta} = 0.42$ HU/SD, 95% CI $[-0.05,\ 0.88]$, $p = 0.079$ in the combined model.

Key result. The PRS does not improve model fit over clinical variables alone (ΔAIC = +0.8, far from the -2 threshold). The PRS-only model is substantially worse than clinical alone (ΔAIC = +33).

Table 2 — Causal Mediation

What we are looking at. MAL causes emphysema progression. But does it do so *through* the blood proteins, or through a completely separate path?

- **NDE (Natural Direct Effect) = 19.7 HU, $p = 0.004$:** the portion of MAL's effect on progression that does *not* go through the protein score. This is large and significant — meaning MAL strongly drives progression through mechanisms independent of what we measured in blood. A one-unit increase in MAL is associated with 19.7 HU more progression via the direct path alone.
- **NIE (Natural Indirect Effect) = -0.12 HU, $p = 0.97$:** the portion of MAL's effect that *does* go through the protein score. This is essentially zero and non-significant — the proteins are not the mechanism by which MAL causes progression.
- **Proportion mediated = -0.6%:** of the total MAL effect on progression, only -0.6% is explained by the blood protein pathway. Negative proportion can occur when the indirect effect goes in the opposite direction to the total effect; here it just reflects noise around zero.
- **Total effect = 19.6 HU:** the sum of NDE + NIE. This is the overall association of MAL with progression, and it is almost entirely direct.
- **In plain terms:** MAL damages the lung directly. The protein changes we detect in blood are a reflection of that process, not a driver of it.

How it was calculated. The pathway tested is: MAL → Proteomic Risk Score → Emphysema Progression. Fitted using the `medflex` R package (natural effects framework) with robust sandwich standard errors.

The idea is to decompose the total effect of MAL on progression into:

- **NDE** — how much of the MAL effect operates through pathways *other than* the protein score (the direct path)
- **NIE** — how much operates *through* the protein score (the mediated path)

Two models are fitted internally by `medflex`:

$$\text{Imputation model: } \Delta\rho \ \sim \ \texttt{meanmal} \times \text{PRS} + \text{covariates}$$

$$\text{Natural effects model: } \Delta\rho \ \sim \ \texttt{meanmal}_0 + \texttt{meanmal}_1 + \text{covariates}$$

$\texttt{meanmal}_0$ is the MAL value at which the mediator (PRS) is set (direct path); $\texttt{meanmal}_1$ is the actual MAL value driving the outcome (indirect path). The difference between them isolates the mediated portion.

Table 2. Natural effects model results.

Effect	Estimate (HU)	z	p
Natural Direct Effect (NDE)	+19.72	2.92	0.004
Natural Indirect Effect (NIE)	−0.12	−0.04	0.971
Total effect	+19.60	—	—
Proportion mediated	−0.6%		

Key result. MAL has a large, significant direct effect on emphysema progression (NDE $p = 0.004$). The proteomic risk score mediates essentially none of this effect (NIE $p = 0.97$, proportion $= -0.6\%$).

Supplement

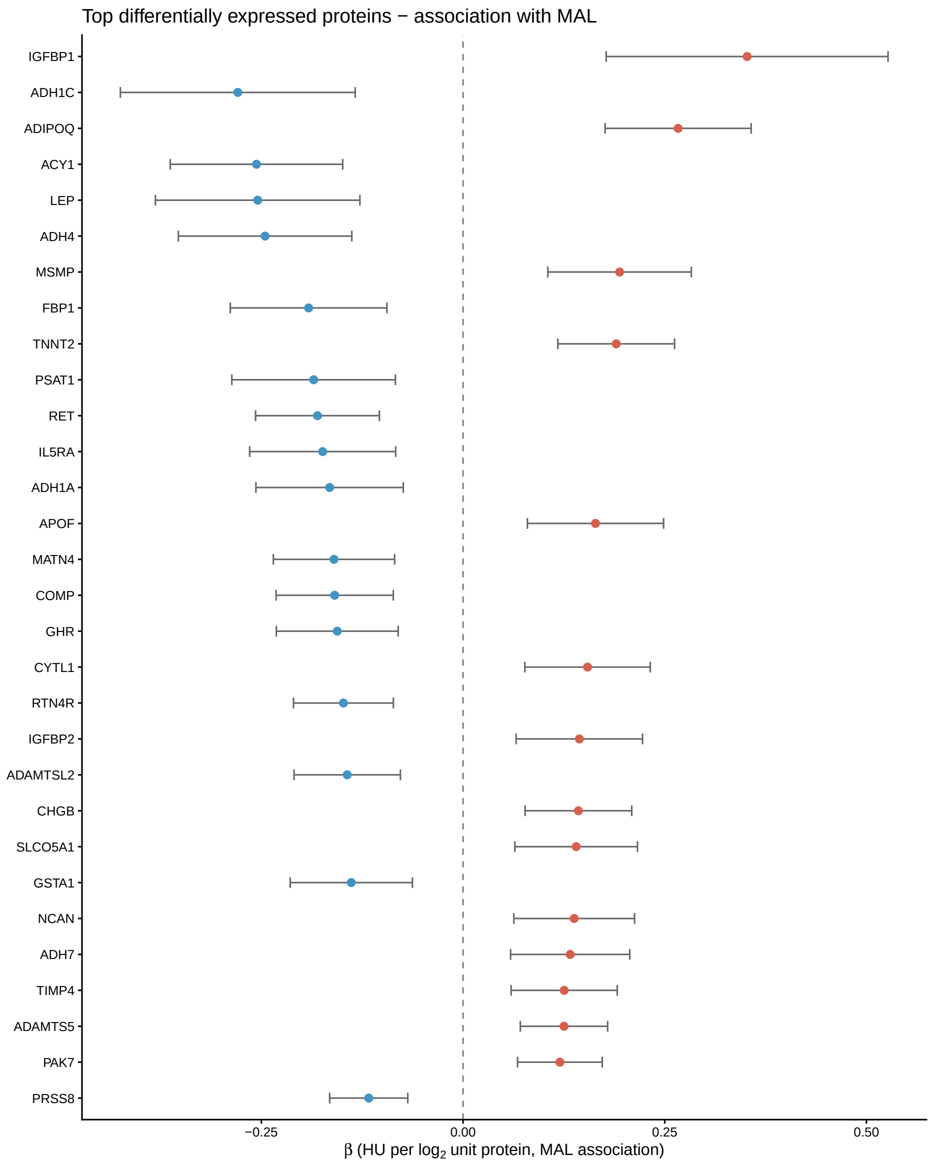
Supplementary Tables

- **S1 — All differential expression results** (4,979 proteins): gene, target, $\hat{\beta}$, SE, t, p, FDR.
  File: `MAL_omics/mal_de_results.csv`
- **S2 — PRS weights** (378 proteins): gene, target, $\hat{\beta}_{\text{OLS}}$, 95% CI, p.
  File: `MAL_omics/mal_risk_score_weights.csv`
- **S3 — Per-protein mediation** (378 proteins in the PRS, individual mediation for each): NDE, NIE, total, proportion mediated, NIE p.
  File: `MAL_omics/mal_mediation_supp.csv`

  Top 6 individual mediators (NIE $p < 0.05$): TREM2 (13%, $p = 0.017$), ADAMTS5 (5.7%, $p = 0.021$), SEZ6L (17%, $p = 0.025$), RSPO1 (14%, $p = 0.038$), GREM2 (12%, $p = 0.039$), ADH7 (7%, $p = 0.044$). All small in absolute magnitude.

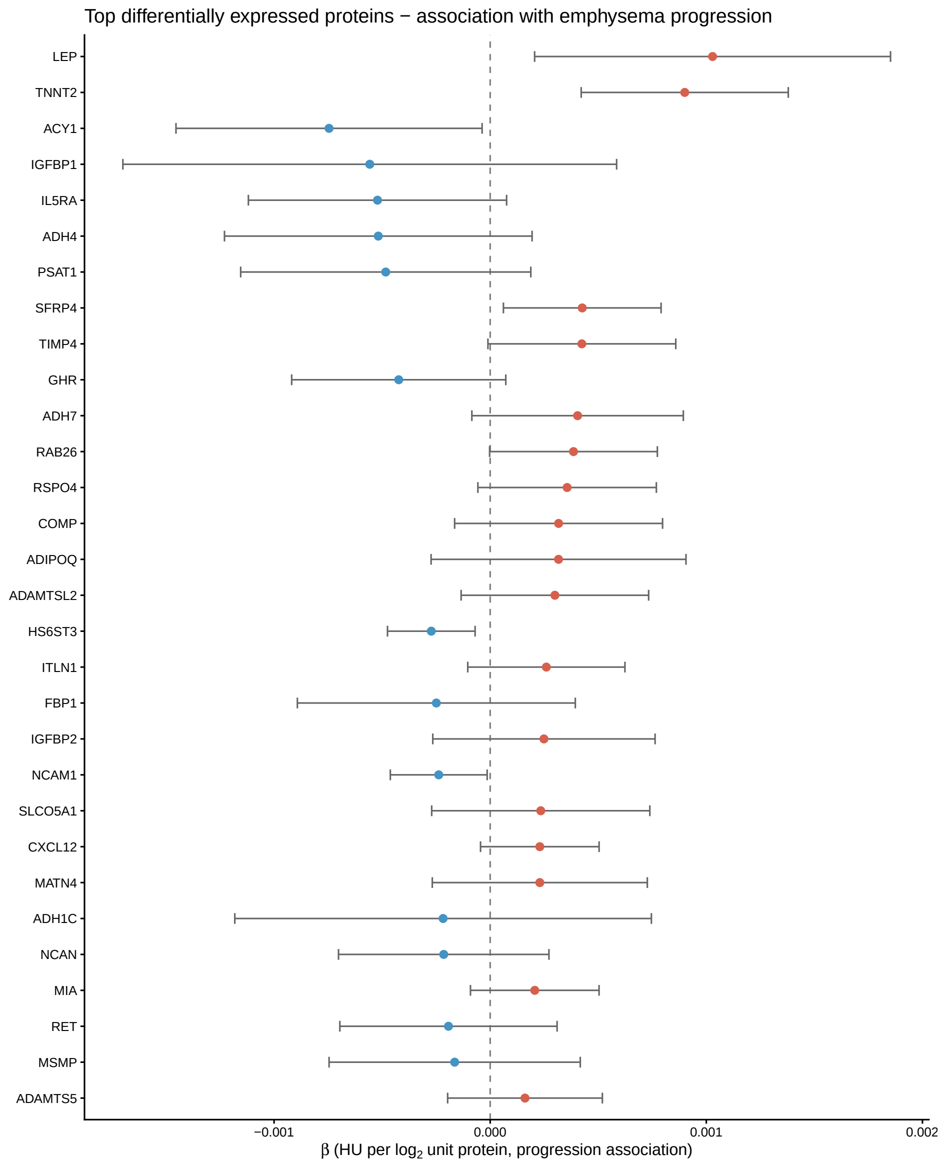
Supplementary Figures

- **Supp Figure A — Forest plot: proteins $\sim$ MAL.** Top 30 FDR-significant proteins, $\hat{\beta}$ for MAL association with 95% CI.



Supplementary Figure A. Forest plot of top 30 differentially expressed proteins — association with MAL.

- **Supp Figure B — Forest plot: proteins $\sim$ emphysema progression.** Same 30 proteins, $\hat{\beta}$ for progression association (adjusted for baseline lung density and covariates).



Supplementary Figure B. Forest plot of top 30 differentially expressed proteins — association with emphysema progression.

What Was Not Found / Assumptions Made

1. **FEV_1% predicted**: column `FEV1pp_P2` does not exist in the phenotype file. Not included in Table 1 or any model. Check the data dictionary for the correct column name.
2. **BMI**: no BMI column found. Not included in Table 1 or the clinical regression model. The clinical model therefore adjusts for age, sex, race, smoking, and pack-years only. AIC comparisons may change once BMI is added.
3. **Figure 1 (study schematic)**: not automatically generated. Needs to be drawn manually.
4. **STRING network PDF**: only PNG was saved. The `STRINGdb` R package does not support direct PDF output via `plot_network()`; a PNG at 150 dpi was used instead.

5. **Table 1 values**: the exact means and percentages are in the notebook HTML output (cell 9) but could not be embedded here because the data are PHI and not committed. The LaTeX rows are placeholders.